04_02: RQ1 – Dataset Distribution and Source Grouping Matrix

1. **Objective**

This document addresses the first component of **Research Question 1**: *" What are the characteristics of the datasets commonly used for machine learning-based insider threat detection?"* It provides a systematic grouping of the 110 primary studies based on their data provenance, distinguishing between standardized benchmarks, real-world logs, and multi-source fusion approaches.

2. Dataset Distribution Summary (Full Breakdown)

The following table provides the quantitative breakdown of the dataset grouping types identified in the literature, maintaining the specific sub-type details for each category.

Dataset Grouping Type	Dataset Family / Sub-type	Total Count	Percentage (%)
Single Source	CERT (r4.1–r6.2)	67	60.91%
	Real-world / Proprietary: Enterprise	4	3.64%
	Real-world / Proprietary: Healthcare	2	1.82%
	Real-world / Proprietary: Telemetry/Network	2	1.82%
	Real-world / Proprietary: Military/Org	2	1.82%
	Specialized Synthetic / Org logs	4	3.64%
	Enron email corpus	2	1.82%
	UEBA and Ontology testbeds	2	1.82%
	Special Repository / Biometric	2	1.82%
Multi-Source Fusion	CERT + LANL + TWOS	4	3.64%
	CERT + Network/IDS (NSL-KDD, etc.)	3	2.73%
	Enron + Other (TWOS, CERT, etc.)	3	2.73%
	Biometric + Log/Simulated data	3	2.73%
	CERT + Specialized (Wiki, CCTV, etc.)	4	3.64%
	IoT / Industrial + Network logs	3	3.64%
	Other Multi-source / UEBA / Synthetic	3	3.64%
Total		**110**	**100%**

The following table provides the dataset used in every one of the 110 studies before grouping

#	Author (Year)	Dataset Source Grouping	Dataset Name(s)
1	**Adun (2023)**	Single Source	CERT (Kaggle v1)
2	**Ahmad (2025)**	Single Source	Real-world Production Logs (Wazuh/Windows)
3	**Ahmadi (2025)**	Single Source	Anonymized Enterprise Logs
4	**Alabdulkreem (2022)**	Single Source	Aggregated Internal Logs (Logon, Email, etc.)
5	**Ali (2025)**	Single Source	US Army Insider Threat Hub (Narrative Text)
6	**Al-Mhiqani (2021)**	Single Source	CERT r4.2

7	**Al-Mhiqani (2022)**	Single Source	CERT r4.2
8	**Almusawy (2024)**	Single Source	CERT r4.2
9	**Al-Shehari and Alsawail (2021)** **One-Hot Encoding**	Single Source	CERT r4.2
10	**Al-shehri Abdullah (2022)** Multi-Sequence Representation	Single Source	CERT r4.2 (32M events)
11	**Al-Shehari (2023) [IF]**	Single Source	CERT r4.2
12	**Al-Shehari (2023) [Resampling]**	Single Source	CERT r4.2
13	**Al-Shehari (2024) [CNN]**	Single Source	CERT r4.2
14	**Al-Shehari (2024) [LOF]**	Single Source	CERT r4.2
15	**AlSlaiman (2023)**	Single Source	CERT r4.2
16	**Amiri-Zarandi (2023)**	Single Source	CERT r4.2
17	**Amuda (2022)**	Single Source	CERT v4.2
18	**Anakath (2022)**	**Multi-Source**	Open-source Biometric + Simulated Logs
19	**Anju (2024)**	Single Source	CERT r4.2 & r6.2
20	**Asha (2024)**	Single Source	CERT (CMU) v3.2
21	**Budžys (2024)**	**Multi-Source**	CMU Keystroke + GREYC-NISLAB
22	**Cai (2024)**	Single Source	CERT r4.2 & r5.2
23	**Ding (2024)**	Single Source	CERT v4.2
24	**Dong (2025)**	Single Source	CERT r5.2 & r6.2
25	**Dosh (2021)**	Single Source	CERT r4.1 & r4.2
26	**Duan (2024)**	Single Source	CMU-CERT r4.2
27	**Erola (2022)**	Single Source	Real-world Enterprise Logs (3 Companies)
28	**Fei (2023)**	**Multi-Source**	CERT r4.2 + LANL
29	**Feng (2025)**	Single Source	CERT r4.2
30	**Ferraro (2025)**	**Multi-Source**	PicoDomain + CERT r5.2
31	**Gayathri (2024)**	Single Source	CERT r4.2 & r5.2
32	**Gayathri R (2025)**	Single Source	CERT v4.2 & v5.2
33	**Gayathri B (2025)**	**Multi-Source**	CERT + NSL-KDD + UNSW-NB15
34	**Gonzales (2025)**	**Multi-Source**	Synthetic Action Databases + UniProt
35	**Greitzer (2021)**	Single Source	SOFIT Ontology Testbed
36	**Rahman (2022)**	Single Source	CERT r4.2
37	**Haq (2022)**	Single Source	Enron Email
38	**He (2022)**	**Multi-Source**	Enron + monkey.org + CERT r6.2
39	**He (2024)**	**Multi-Source**	Enron + monkey.org + CERT r6.2
40	**Hong (2023)**	Single Source	CERT r4.2
41	**Huang (2025)**	**Multi-Source**	KDD-UEBA + NSL-UEBA + CIC-UEBA + KDD-UNBLogs
42	**Hurst (2022)**	Single Source	UK Hospital EHR Logs

43	**Jaiswal (2024)**	Single Source	CERT r4.2 & r5.2
44	**Janjua (2021)**	**Multi-Source**	Enron + TWOS
45	**Jyosthna (2021)**	Single Source	CERT r6.1
46	**Kabir (2023)**	Single Source	CERT r4.2
47	**Kamatchi (2025)**	**Multi-Source**	Simulated (NS-3) + X-IIoTID
48	**Kan (2023)**	Single Source	CERT r4.2
49	**Kim (2023)**	Single Source	CERT (Scenarios 1-3)
50	**Kong (2025)**	**Multi-Source**	CERT r4.2 + SEA + HCT + PU
51	**Kotb (2025)**	Single Source	CERT r4.2 (Full)
52	**Lavanya (2024)**	**Multi-Source**	CERT r6.1 & r6.2 and IoT enabled institutional log records
53	**Le (2021) [Exploring]**	Single Source	CERT r4.1, r4.2, r5.1, r5.2, r6.2
54	**Le (2021) [Ensembles]**	**Multi-Source**	CERT + LANL + TWOS
55	**Li Dongyang (2021)**	Single Source	CERT r4.2 & r6.2
56	**Li Ximing (2023)**	Single Source	CERT r4.2 & r6.2
57	**Li Ximing (2024)**	**Multi-Source**	CERT + UMDWikipedia
58	**Lilhore et al. (2025)**	**Multi-Source**	CIC-IoT 2023 + UNSW-NB15
59	**Liu et al. (2025)**	**Multi-Source**	CERT + DARPA1999
60	**Manoharan (2024)**	Single Source	CERT r4.2
61	**Matos et al. (2025)**	Single Source	Synthetic Healthcare Logs
62	**Medvedev et al. (2025)**	**Multi-Source**	CMU + KeyRecs
63	**Mehmood et al. (2023)**	Single Source	CERT (Customized)
64	**Mehnaz (2021)**	Single Source	Wikipedia File Repository Logs
65	**Mladenovic (2024)**	Single Source	CERT r4.2
66	**Nasir et al. (2021)**	Single Source	CERT r4.2
67	**Nikiforova (2024)**	Single Source	Real-world Audit Logs (Proprietary)
68	**Pal et al. (2023)**	Single Source	CERT r4.2, r5.2, r6.2
69	**Patel (2025)**	Single Source	Landauer UEBA Log Dataset
70	**Peccatiello (2023)**	Single Source	CERT v4.2
71	**Pérez-Gomariz (2025)**	Single Source	Fortune 500 Incident Logs
72	**Prasad (2024) [Enhanced]**	Single Source	CERT v5.2
73	**Prasad (2025) [Hybrid]**	Single Source	CERT
74	**Prasad (2025) [Deep]**	Single Source	CERT
75	**Qawasmeh (2025)**	Single Source	Synthetic Organizational Dataset
76	**Qi et al. (2025)**	**Multi-Source**	CERT r4.2 + LANL
77	**Racherache (2023)**	Single Source	Real-world Network Traffic (Industry Partners)
78	**Randive (2023a)**	Single Source	CERT r4.2
79	**Randive (2023b)**	Single Source	CERT r4.2 & r5.2
80	**Rauf et al. (2021)**	Single Source	CERT v4.2
81	**Saminathan (2023)**	Single Source	Simulated Organizational Environment
82	**Bin Sarhan (2023)**	Single Source	CERT r4.2
83	**Senevirathna (2025)**	**Multi-Source**	CERT + Video CCTV Data

84	**Singh et al. (2022)**	Single Source	CERT v4.2
85	**Singh et al. (2023)**	Single Source	CERT r4.2
86	**Sivakrishna (2025)**	Single Source	CERT r4.2, r5.2, r6.2
87	**Song et al. (2024)**	Single Source	CERT v6.2
88	**Tabassum (2024)**	Single Source	UK Hospital EHR Logs
89	**Tao (2023)**	Single Source	SapiMouse
90	**Tao (2025)**	Single Source	CERT v4.2
91	**Tian Zhihong (2024)**	Single Source	CERT v6.2
92	**Tian Tian (2025)**	Single Source	CERT v6.2
93	**Villarreal-Vasquez (2023)**	Single Source	Commercial EDR System Logs
94	**Wall (2021)**	Single Source	CERT r4.1
95	**Wang Cheng and Zhu (2022)**	**Multi-Source**	CERT + Banking + Kyoto + Darknet + Gowalla
96	**Wang & El Saddik (2023)**	Single Source	CERT r4.2 & r6.2
97	**Wang Jiarong et al deep (2023)**	Single Source	CERT r4.2
98	**Wang et al Fusion(2022)**	Single Source	CERT r4.2 (HTTP)
99	**Wang (2024)**	Single Source	CERT r6.2
100	**Wei Yichen (2021)**	**Multi-Source**	CERT r6.2 + KDD CUP 99
101	**Wei Zhiyuan (2024)**	Single Source	CERT r4.2
102	**Wen et al. (2023)**	Single Source	Enron Email
103	**Xiao Junchao (2023)**	Single Source	CERT r6.2
104	**Xiao Haitao (2024)**	Single Source	CERT
105	**Xiao Fengrui (2025)**	**Multi-Source**	LANL + CERT r6.2 + TWOS
106	**Ye Xiaoyun (2025)**	Single Source	CERT v4.2 & v5.2
107	**Yi & Tian (2024)**	Single Source	CERT r4.2
108	**Yumlembam (2025)**	Single Source	CERT r5.2 & r6.2
109	**Zhang et al. (2021)**	Single Source	CERT v4.2 & v6.2
110	**Zhu et al. (2024)**	Single Source	CERT

The use of datasets can be grouped into 4 groups as shown in the tables below:

Dataset Grouping Type	Count	Percentage (%)	Representative Study Examples
Single Source: CERT Benchmark (Standard versions: r4.1, r4.2, r5.2, r6.2)	**67**	**60.91%**	Adun (2023), Al-Mhiqani (2021), Al-Shehari (2023), Nasir (2021), Randive (2023a)
Multi-Source Fusion (2–5+ Distinct Datasets) (Combining benchmarks or cross-domain sources)	**23**	**20.91%**	**CERT + LANL:** Fei (2023), Qi (2025); **Enron + TWOS:** Janjua (2021); **CERT + NSL-KDD:** Gayathri B (2025)
Single Source: Real-world/Proprietary Logs	9	8.18%	**Enterprise:** Ahmad (2025), Erola (2022); **Hospital/EHR:**

(Production logs, EDR telemetry, EHR audit logs)			Hurst (2022), Tabassum (2024); **EDR:** Villarreal-Vasquez (2023)
Single Source: Specialized Synthetic (Niche frameworks like SOFIT or Army InT Hub)	8	7.27%	Ali (2025), Greitzer (2021), Qawasmeh (2025), Alabdulkreem (2022)
Single Source: Enron Email Corpus (Focusing exclusively on corporate communication)	2	1.82%	Haq (2022), Wen et al. (2023)
Single Source: Biometric Dynamics (Focusing exclusively on Mouse/Keystroke logs)	1	0.91%	Tao (2023)
Total	**110**	**100%**	

Dataset Grouping Type	Total Count	Percentage (%)
Single Source: CERT Benchmark	**67**	**60.91%**
Multi-Source Fusion (2–5+ Datasets)	**23**	**20.91%**
Single Source: Real-world / Proprietary	**10**	**9.09%**
Single Source: Other / Specialized Synthetic	**10**	**9.09%**
Total	**110**	**100%**

Group 1: Single Source: CERT Benchmark (n=67)

These studies utilize the CMU-CERT synthetic dataset exclusively (versions r4.1 to r6.2).

#	Author (Year)	#	Author (Year)	#	Author (Year)
1	Adun (2023)	**24**	Gayathri R (2025)	**47**	Rauf et al. (2021)
2	Al-Mhiqani (2021)	**25**	Hong (2023)	**48**	Singh et al. (2022)
3	Al-Mhiqani (2022)	**26**	Jaiswal (2024)	**49**	Singh et al. (2023)
4	Almusawy (2024)	**27**	Jyosthna (2021)	**50**	Sivakrishna (2025)
5	Al-Shehari & Alsawail (2021)	**28**	Kabir (2023)	**51**	Song et al. (2024)
6	Al-shehri Abdullah (2022)	**29**	Kan (2023)	**52**	Tao (2025)
7	Al-Shehari (2023) [IF]	**30**	Kim (2023)	**53**	Tian Zhihong (2024)
8	Al-Shehari (2023) [Res]	**31**	Kotb (2025)	**54**	Tian Tian (2025)
9	Al-Shehari (2024) [CNN]	**32**	Le (2021) [Exploring]	**55**	Wall (2021)
10	Al-Shehari (2024) [LOF]	**33**	Li Dongyang (2021)	**56**	Wang & El Saddik (2023)

11	AlSlaiman (2023)	**34**	Li Ximing (2023)	**57**	Wang Jiarong deep cluster (2023)
12	Amiri-Zarandi (2023)	**35**	Manoharan (2024)	**58**	Wang et al. Fusion (2022)
13	Amuda (2022)	**36**	Mehmood et al. (2023)	**59**	Wang (2024)
14	Anju (2024)	**37**	Mladenovic (2024)	**60**	Wei Zhiyuan (2024)
15	Asha (2024)	**38**	Nasir et al. (2021)	**61**	Xiao Junchao (2023)
16	Bin Sarhan (2023)	**39**	Pal et al. (2023)	**62**	Xiao Haitao (2024)
17	Cai (2024)	**40**	Peccatiello (2023)	**63**	Ye Xiaoyun (2025)
18	Ding (2024)	**41**	Prasad (2024) [Enhanced]	**64**	Yi & Tian (2024)
19	Dong (2025)	**42**	Prasad (2025) [Hybrid]	**65**	Yumlembam (2025)
20	Dosh (2021)	**43**	Prasad (2025) [Deep]	**66**	Zhang et al. (2021)
21	Duan (2024)	**44**	Rahman (2022)	**67**	Zhu et al. (2024)
22	Feng (2025)	**45**	Randive (2023a)		
23	Gayathri (2024)	**46**	Randive (2023b)		

Group 2: Multi-Source Fusion (n=23)

These studies combine fundamentally different data sources (e.g., system logs + network traffic + text) to build a robust feature space.

#	Author (Year)	Specific Datasets Combined
1	**Anakath (2022)**	Open-source Biometric + Simulated Logs
2	**Budžys (2024)**	CMU Keystroke + GREYC-NISLAB
3	**Fei (2023)**	CERT r4.2 + LANL
4	**Ferraro (2025)**	PicoDomain + CERT r5.2
5	**Gayathri B (2025)**	CERT + NSL-KDD + UNSW-NB15
6	**Gonzales (2025)**	Synthetic Action Databases + UniProt
7	**He (2022)**	Enron + monkey.org + CERT r6.2
8	**He (2024)**	Enron + monkey.org + CERT r6.2
9	**Huang (2025)**	KDD-UEBA + NSL-UEBA + CIC-UEBA + KDD-UNBLogs
10	**Janjua (2021)**	Enron + TWOS
11	**Kamatchi (2025)**	Simulated (NS-3) + X-IIoTID
12	**Kong (2025)**	CERT r4.2 + SEA + HCT + PU
13	**Lavanya (2024)**	CERT r6.1 & r6.2 and IoT enabled institutional log records
14	**Le (2021) [Ensembles]**	CERT + LANL + TWOS
15	**Li Ximing (2024)**	CERT + UMDWikipedia
16	**Lilhore et al. (2025)**	CIC-IoT 2023 + UNSW-NB15
17	**Liu et al. (2025)**	CERT + DARPA1999
18	**Medvedev et al. (2025)**	CMU + KeyRecs
19	**Qi et al. (2025)**	CERT r4.2 + LANL

20	**Senevirathna (2025)**	CERT + Video CCTV
21	**Wang Cheng (2022)**	CERT + Banking + Kyoto + Darknet + Gowalla
22	**Wei Yichen (2021)**	CERT r6.2 + KDD CUP 99
23	**Xiao Fengrui (2025)**	LANL + CERT r6.2 + TWOS

Group 3: Single Source: Real-world / Proprietary (n=10)

These studies use internal enterprise logs or telemetry from live operational environments.

#	Author (Year)	Source Type
1	**Ahmad (2025)**	Real-world Production Logs (Windows/Wazuh)
2	**Ahmadi (2025)**	Anonymized Corporate Logs
3	**Ali (2025)**	US Army Insider Threat Hub (Narrative)
4	**Erola (2022)**	Enterprise Logs (Proxy, VPN, AD from 3 companies)
5	**Hurst (2022)**	UK Hospital EHR Logs
6	**Nikiforova (2024)**	Proprietary Business Audit Logs
7	**Pérez-Gomariz (2025)**	Fortune 500 Incident Logs
8	**Tabassum (2024)**	UK Hospital EHR Logs
9	**Villarreal-Vasquez (2023)**	Commercial EDR System Telemetry
10	**Racherache (2023)**	Real-world Network Traffic

Group 4: Single Source: Other / Specialized Synthetic (n=10)

These studies use specific domain datasets like Enron, Biometrics, or specialized UEBA benchmarks.

#	Author (Year)	Specialized Source
1	**Alabdulkreem (2022)**	Custom Aggregated Internal Logs
2	**Greitzer (2021)**	SOFIT Ontology Testbed
3	**Haq (2022)**	Enron Email Corpus
4	**Matos et al. (2025)**	Synthetic Healthcare Access Logs
5	**Mehnaz (2021)**	Wikipedia File Repository Logs
6	**Patel (2025)**	Landauer UEBA Log Dataset
7	**Qawasmeh (2025)**	Synthetic Organizational Dataset
8	**Saminathan (2023)**	Simulated Organizational Logs
9	**Wen et al. (2023)**	Enron Email Corpus
10	**Tao (2023)**	SapiMouse (Biometric)